

Domain Controller and Active Directory Lab

Part 5 – Establishing Remote Access Server (RAS) and Network Address Translation (NAT)

Author: Denzel Corson

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Technologies Used

- **Oracle VirtualBox**
- **Windows Server 2022**
- **Server Manager**
- **Active Directory Domain Services**

OBJECTIVE:

The objective of this lab is to establish a Remote Access Server (RAS) and Network Address Translation (NAT). Creating RAS & NAT will allow Window clients, attached to the internal enterprise network to access external networks through the Domain Controller's external network connection.

Lab Setup:

This lab continues from Part 4, where a dedicated Domain Administrator account was created for managing the enterprise domain. At this point, the Domain Controller has been promoted, Active Directory Domain Services is running, and separate internal and external network adapters have already been configured.

For this part of the lab, Remote Access and Routing services were installed on the Domain Controller to establish Network Address Translation (NAT). NAT will allow future Windows clients connected to the private internal network to access external networks through the Domain Controller's external network connection.

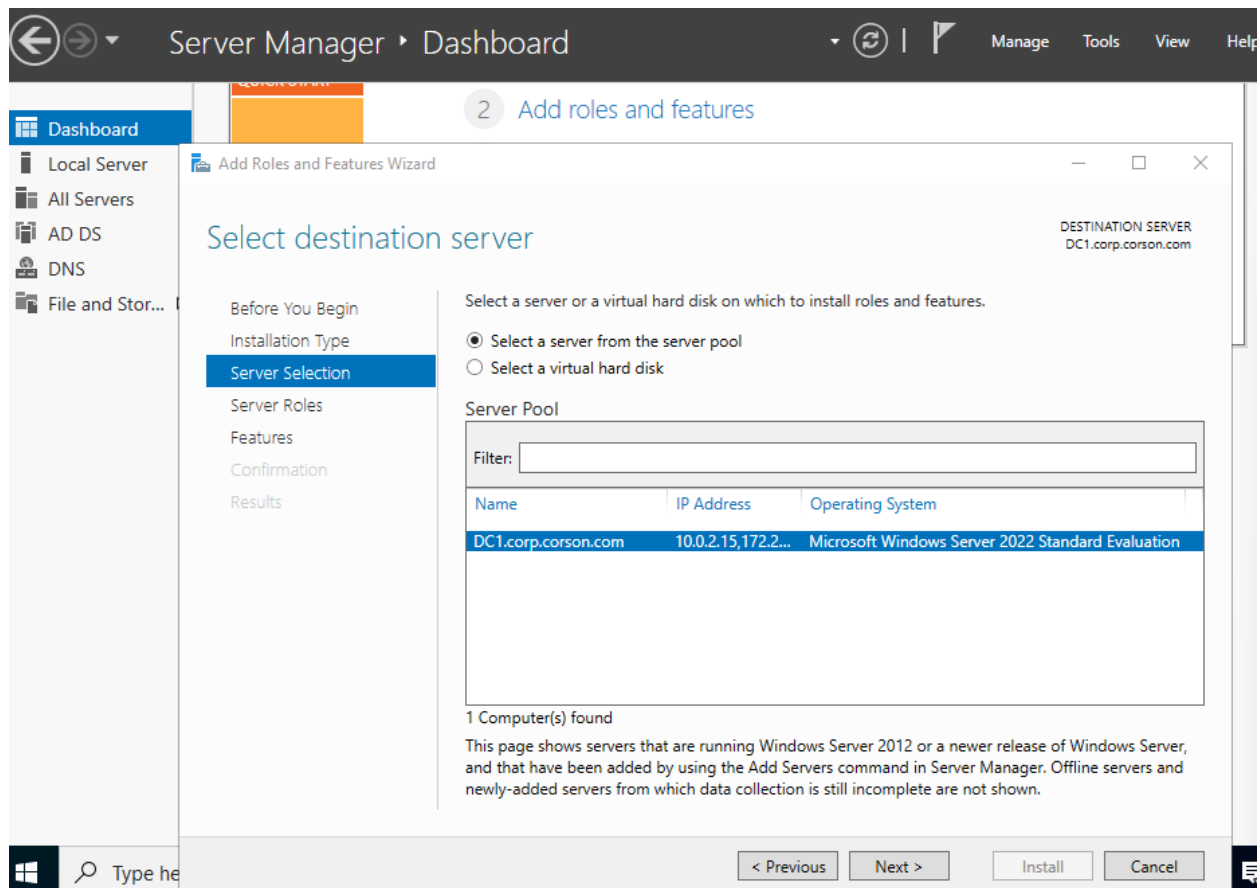


Figure 1. Using Server Manager, the **Add Roles and Features Wizard** was opened and DC1.corp.corson.com was selected as the destination server. The **Remote Access** role was installed along with DirectAccess and VPN (RAS) and Routing services. These services provide the Domain Controller with the components needed to configure Routing and Remote Access Services (RRAS).

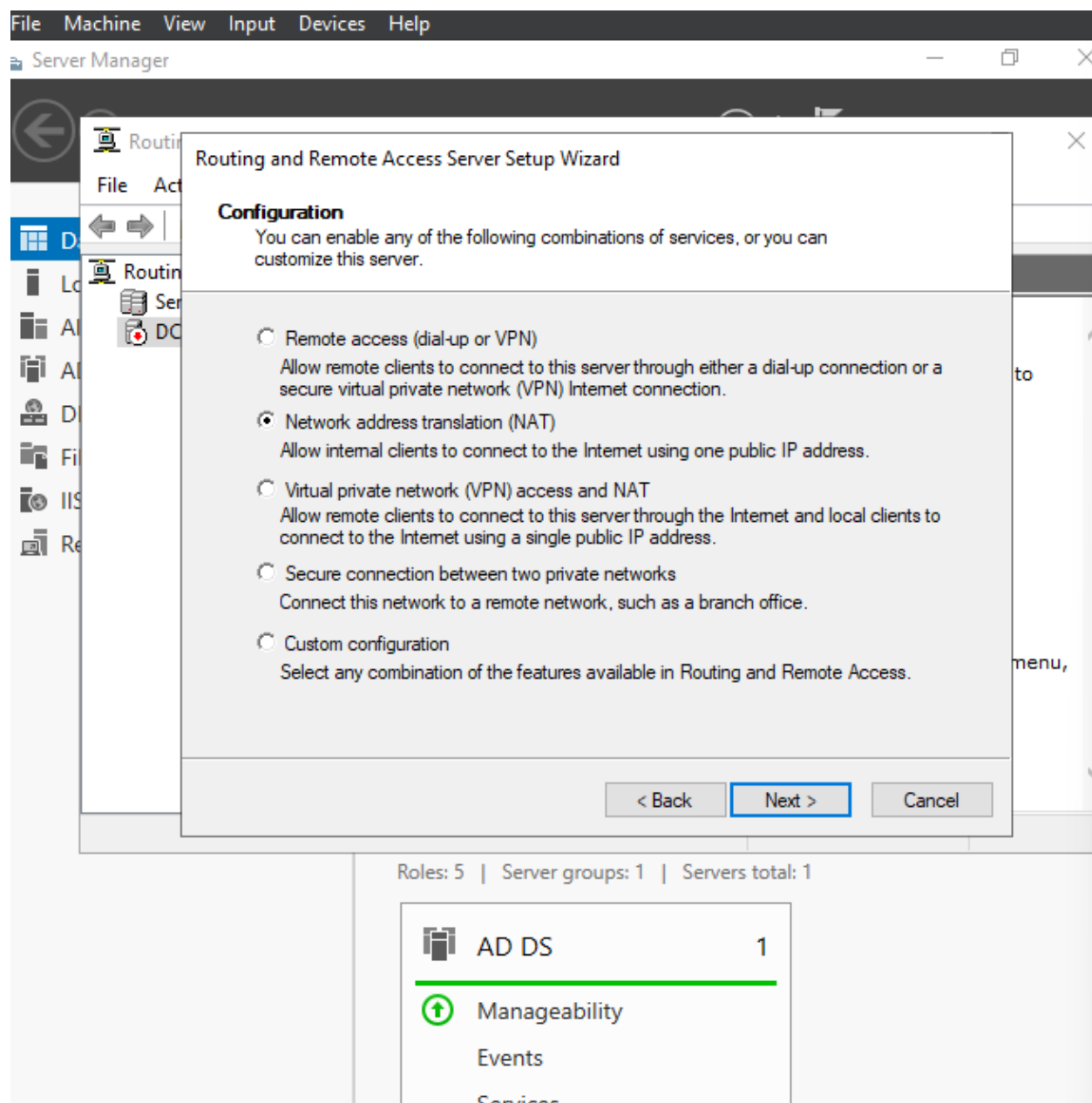


Figure 2. After the Remote Access and Routing services were installed, the Routing and Remote Access Server Setup Wizard was opened. Network Address Translation (NAT) was selected. This allows clients located on the private internal network to communicate with external networks using the Domain Controller's external network.

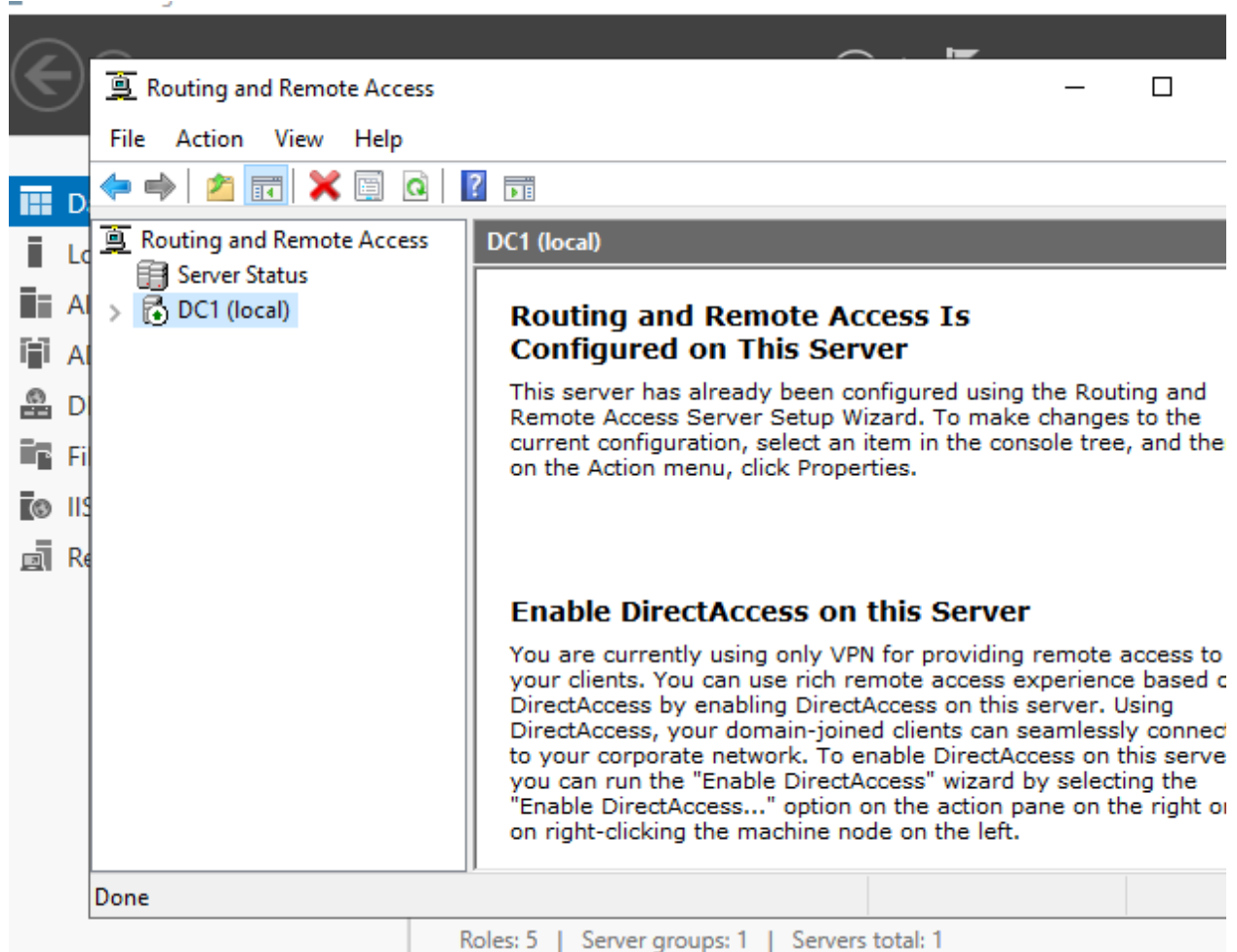


Figure 3. After completing the setup wizard, Routing and Remote Access and NAT were successfully configured on DC1. DC1 can now route traffic from the internal network through its external network interface.

What Changed?

Before this configuration, the Domain Controller had two separate network connections: an external adapter providing access outside of the lab environment and an internal adapter used for communication within the private enterprise network. However, clients connected only to the internal network did not have a route to external networks.

Installing Routing and Remote Access Services and configuring NAT allows the DC to act as the connection between these two networks, allowing future domain-connected Windows clients to remain on the private enterprise network while still being able to access external network resources.

